Year 7

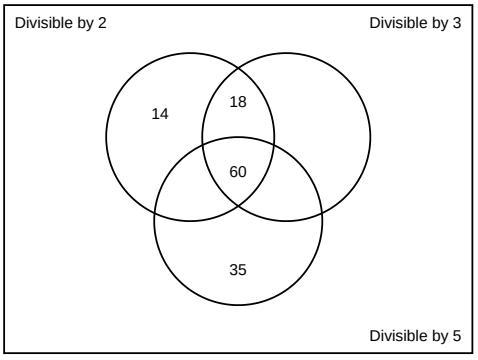
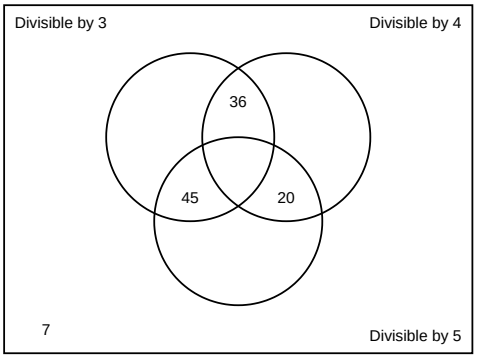
REVISION PACK

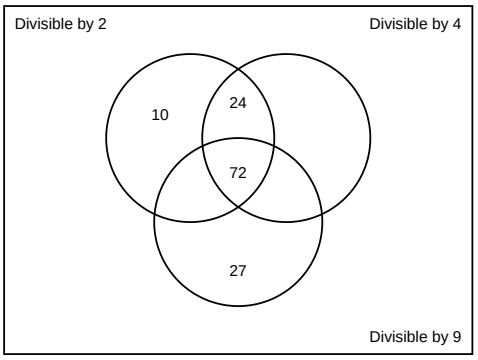
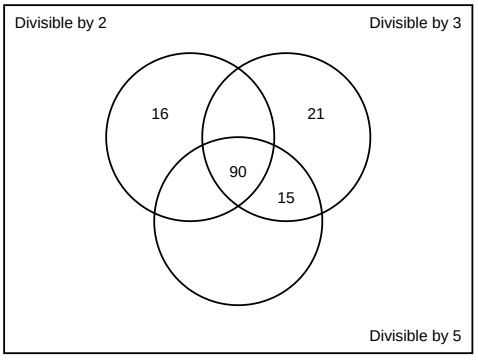
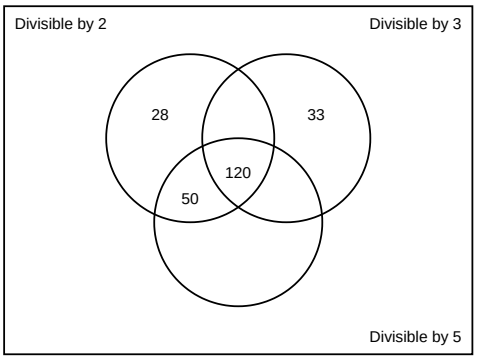
Unit 2

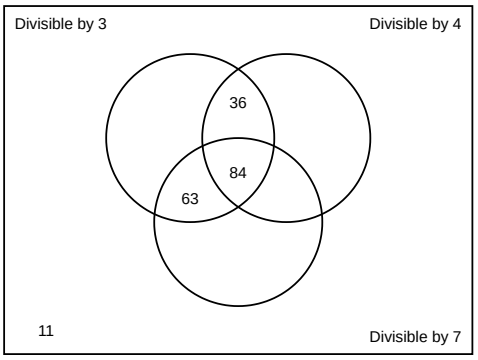
ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

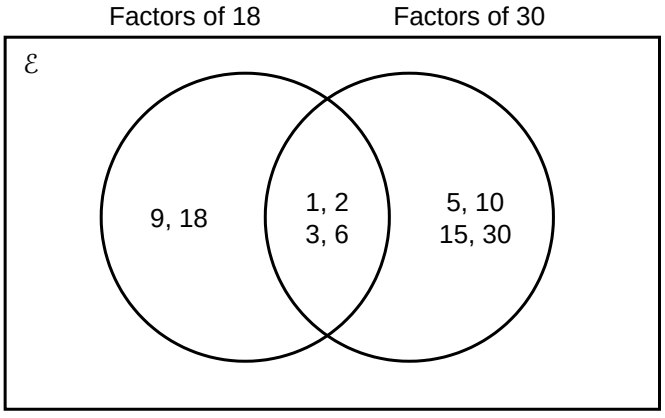
Unit 2 · Venn diagrams

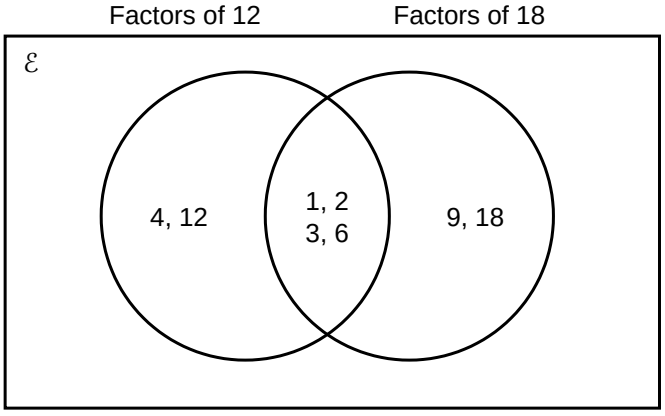
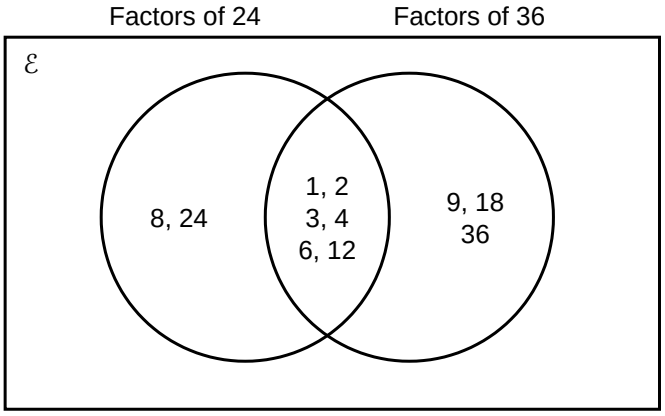
Q	Working	Answer
sorting numbers in a Venn diagram		
1.1	$60:$ divisible by $2, 3, 5$ $18:$ divisible by $2, 3$ $35:$ divisible by 5 $14:$ divisible by 2 	**60 in all three; 18 in 2 and 3; 35 in only 5; 14 in only 2**
1.2	$45:$ divisible by $3, 5$ $20:$ divisible by $4, 5$ $36:$ divisible by $3, 4$ $7:$ divisible by none 	**45 in 3 and 5; 20 in 4 and 5; 36 in 3 and 4; 7 in outside**

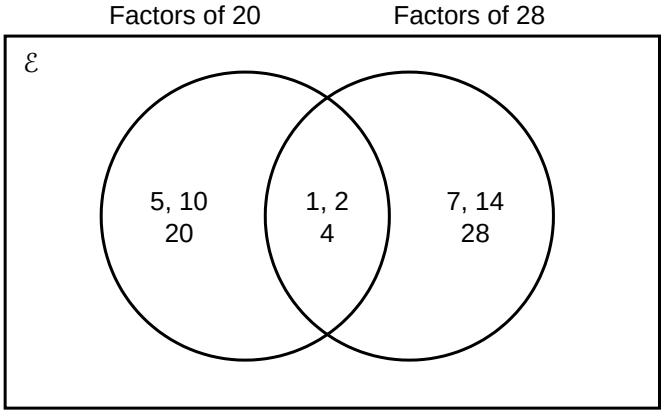
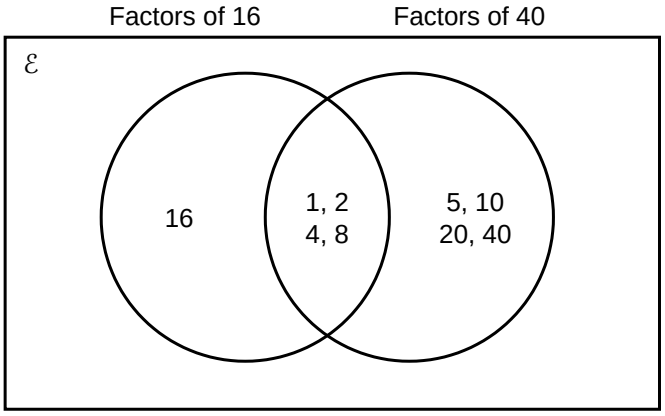
1.3	$72:$ divisible by $2, 4, 9$ $24:$ divisible by $2, 4$ $27:$ divisible by 9 $10:$ divisible by 2 	**72 in all three; 24 in 2 and 4; 27 in only 9; 10 in only 2**
1.4	$90:$ divisible by $2, 3, 5$ $15:$ divisible by $3, 5$ $16:$ divisible by 2 $21:$ divisible by 3 	**90 in all three; 15 in 3 and 5; 16 in only 2; 21 in only 3**
1.5	$120:$ divisible by $2, 3, 5$ $28:$ divisible by 2 $33:$ divisible by 3 $50:$ divisible by $2, 5$ 	**120 in all three; 28 in only 2; 33 in only 3; 50 in 2 and 5**

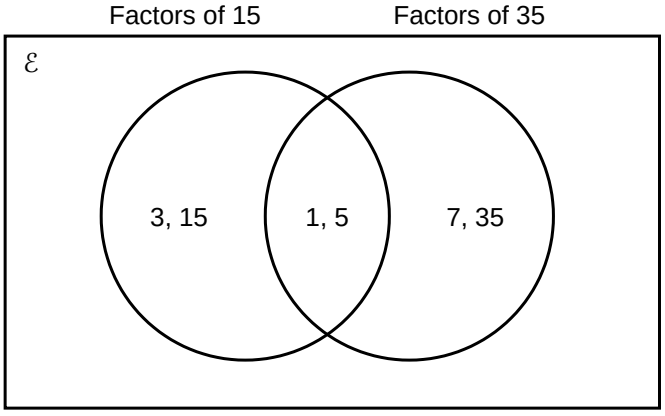
1.6	$84:$ divisible by $3, 4, 7$ $36:$ divisible by $3, 4$ $63:$ divisible by $3, 7$ $11:$ divisible by none 	**84 in all three; 36 in 3 and 4; 63 in 3 and 7; 11 in outside**

Unit 2 · Factors, HCF and LCM

Q	Working	Answer
factors in a Venn diagram, HCF and LCM		
2.1	**(a)** factors of 18: $1, 2, 3, 6, 9, 18$ factors of 30: $1, 2, 3, 5, 6, 10, 15, 30$ both: $1, 2, 3, 6$ **(b)** biggest number in the middle $= 6$ **(c)** $18: 18, 36, 54, 72, 90$ $30: 30, 60, 90$ 	**(a) Only 18: 9, 18; both: 1, 2, 3, 6; only 30: 5, 10, 15, 30** **(b) 6** **(c) 90**

2.2	**(a)** factors of 12: $1, 2, 3, 4, 6, 12$ factors of 18: $1, 2, 3, 6, 9, 18$ both: $1, 2, 3, 6$ **(b)** biggest number in the middle $= 6$ **(c)** 12 : $12, 24, 36$ 18 : $18, 36$ 	**(a) Only 12: 4, 12; both: 1, 2, 3, 6; only 18: 9, 18** **(b) 6** **(c) 36**
2.3	**(a)** factors of 24: $1, 2, 3, 4, 6, 8, 12, 24$ factors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$ both: $1, 2, 3, 4, 6, 12$ **(b)** biggest number in the middle $= 12$ **(c)** 24 : $24, 48, 72$ 36 : $36, 72$ 	**(a) Only 24: 8, 24; both: 1, 2, 3, 4, 6, 12; only 36: 9, 18, 36** **(b) 12** **(c) 72**

2.4	**(a)** factors of 20: $1, 2, 4, 5, 10, 20$ factors of 28: $1, 2, 4, 7, 14, 28$ both: $1, 2, 4$ **(b)** biggest number in the middle $= 4$ **(c)** $20 : 20, 40, 60, 80, 100, 120, 140$ $28 : 28, 56, 84, 112, 140$ 	**(a) Only 20: 5, 10, 20; both: 1, 2, 4; only 28: 7, 14, 28** **(b) 4** **(c) 140**
2.5	**(a)** factors of 16: $1, 2, 4, 8, 16$ factors of 40: $1, 2, 4, 5, 8, 10, 20, 40$ both: $1, 2, 4, 8$ **(b)** biggest number in the middle $= 8$ **(c)** $16 : 16, 32, 48, 64, 80$ $40 : 40, 80$ 	**(a) Only 16: 16; both: 1, 2, 4, 8; only 40: 5, 10, 20, 40** **(b) 8** **(c) 80**

2.6	**(a)** factors of 15: $1, 3, 5, 15$ factors of 35: $1, 5, 7, 35$ both: $1, 5$ **(b)** biggest number in the middle $= 5$ **(c)** 15 : $15, 30, 45, 60, 75, 90, 105$ 35 : $35, 70, 105$ 	**(a) Only 15: 3, 15; both: 1, 5; only 35: 7, 35** **(b) 5** **(c) 105**

Unit 2 · Negative numbers

Q	Working	Answer
calculating with negative numbers		
3.1	**(a)** $-4 + 9 = 5$ **(b)** $6 \times -7 = -42$ **(c)** $-32 \div -4 = 8$ **(d)** $7 - 15 = -8$ **(e)** $-12 + 15 = 3$	**(a) 5** **(b) -42** **(c) 8** **(d) -8** **(e) 3°C**
3.2	**(a)** $-7 + 3 = -4$ **(b)** $-8 \times 5 = -40$ **(c)** $54 \div -6 = -9$ **(d)** $3 - 11 = -8$ **(e)** $-9 + 13 = 4$	**(a) -4** **(b) -40** **(c) -9** **(d) -8** **(e) 4°C**

3.3	**(a)** $5 + 6 = 11$ **(b)** $-9 \times -4 = 36$ **(c)** $-45 \div 9 = -5$ **(d)** $-2 - 8 = -10$ **(e)** $2 - 10 = -8$	**(a) 11** **(b) 36** **(c) -5** **(d) -10** **(e) -8°C**
3.4	**(a)** $-12 + 5 = -7$ **(b)** $7 \times -7 = -49$ **(c)** $-60 \div -10 = 6$ **(d)** $8 - 20 = -12$ **(e)** $-15 + 9 = -6$	**(a) -7** **(b) -49** **(c) 6** **(d) -12** **(e) -6°C**
3.5	**(a)** $-2 + 10 = 8$ **(b)** $-3 \times -11 = 33$ **(c)** $36 \div -4 = -9$ **(d)** $-6 - 9 = -15$ **(e)** $-18 + 12 = -6$	**(a) 8** **(b) 33** **(c) -9** **(d) -15** **(e) -6°C**
3.6	**(a)** $-9 + 14 = 5$ **(b)** $11 \times -2 = -22$ **(c)** $-81 \div 9 = -9$ **(d)** $12 - 19 = -7$ **(e)** $6 - 14 = -8$	**(a) 5** **(b) -22** **(c) -9** **(d) -7** **(e) -8°C**

Unit 2 · Powers and roots

Q	Working	Answer
squares, cubes, powers of 10 and roots		

4.1	**(a)** $500 \times 500 = 250000$ **(b)** $0.4 \times 0.4 \times 0.4 = 0.064$ **(c)** $10 \times 10 \times 10 \times 10 = 10\,000$ **(d)** $11 \times 11 = 121$ **(e)** $3 \times 3 \times 3 = 27$	**(a) 250000** **(b) 0.064** **(c) 10 000** **(d) 11** **(e) 3**
4.2	**(a)** $800 \times 800 = 640000$ **(b)** $0.3 \times 0.3 \times 0.3 = 0.027$ **(c)** $10 \times 10 \times 10 \times 10 \times 10 \times 10 = 1\,000\,000$ **(d)** $13 \times 13 = 169$ **(e)** $5 \times 5 \times 5 = 125$	**(a) 640000** **(b) 0.027** **(c) 1 000 000** **(d) 13** **(e) 5**
4.3	**(a)** $400 \times 400 = 160000$ **(b)** $0.1 \times 0.1 \times 0.1 = 0.001$ **(c)** $10 \times 10 \times 10 = 1\,000$ **(d)** $9 \times 9 = 81$ **(e)** $10 \times 10 \times 10 = 1000$	**(a) 160000** **(b) 0.001** **(c) 1 000** **(d) 9** **(e) 10**
4.4	**(a)** $900 \times 900 = 810000$ **(b)** $0.6 \times 0.6 \times 0.6 = 0.216$ **(c)** $10 \times 10 \times 10 \times 10 \times 10 = 100\,000$ **(d)** $14 \times 14 = 196$ **(e)** $2 \times 2 \times 2 = 8$	**(a) 810000** **(b) 0.216** **(c) 100 000** **(d) 14** **(e) 2**
4.5	**(a)** $200 \times 200 = 40000$ **(b)** $0.5 \times 0.5 \times 0.5 = 0.125$ **(c)** $10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 10\,000\,000$ **(d)** $15 \times 15 = 225$ **(e)** $7 \times 7 \times 7 = 343$	**(a) 40000** **(b) 0.125** **(c) 10 000 000** **(d) 15** **(e) 7**

4.6	**(a)** $600 \times 600 = 360000$ **(b)** $0.7 \times 0.7 \times 0.7 = 0.343$ **(c)** $10 \times 10 = 100$ **(d)** $10 \times 10 = 100$ **(e)** $6 \times 6 \times 6 = 216$	**(a) 360000** **(b) 0.343** **(c) 100** **(d) 10** **(e) 6**

Unit 2 · Order of operations

Q	Working	Answer
order of operations with powers and roots		
5.1	**(a)** $5^2 = 25$ $6 \times 25 = 150$ **(b)** $81 - 3 \times 4$ $= 81 - 12 = 69$ **(c)** $\sqrt{9} = 3$ $3 + 5 = 8$ $4 \times 8 = 32$ **(d)** $2 \times 3 = 6$ $1 + 6 = 7$ $7^2 = 49$	**(a) 150** **(b) 69** **(c) 32** **(d) 49**
5.2	**(a)** $9^2 = 81$ $2 \times 81 = 162$ **(b)** $64 - 4 \times 5$ $= 64 - 20 = 44$ **(c)** $\sqrt{16} = 4$ $4 + 2 = 6$ $3 \times 6 = 18$ **(d)** $1 \times 5 = 5$ $4 + 5 = 9$ $9^2 = 81$	**(a) 162** **(b) 44** **(c) 18** **(d) 81**

5.3	**(a)** $3^2 = 9$ $5 \times 9 = 45$ **(b)** $49 - 6 \times 2$ $= 49 - 12 = 37$ **(c)** $\sqrt{16} = 4$ $4 + 6 = 10$ $5 \times 10 = 50$ **(d)** $2 \times 4 = 8$ $3 + 8 = 11$ $11^2 = 121$	**(a) 45** **(b) 37** **(c) 50** **(d) 121**
5.4	**(a)** $6^2 = 36$ $4 \times 36 = 144$ **(b)** $100 - 9 \times 9$ $= 100 - 81 = 19$ **(c)** $\sqrt{36} = 6$ $6 + 4 = 10$ $2 \times 10 = 20$ **(d)** $3 \times 3 = 9$ $2 + 9 = 11$ $11^2 = 121$	**(a) 144** **(b) 19** **(c) 20** **(d) 121**
5.5	**(a)** $2^2 = 4$ $7 \times 4 = 28$ **(b)** $36 - 5 \times 3$ $= 36 - 15 = 21$ **(c)** $\sqrt{64} = 8$ $8 + 1 = 9$ $6 \times 9 = 54$ **(d)** $4 \times 2 = 8$ $1 + 8 = 9$ $9^2 = 81$	**(a) 28** **(b) 21** **(c) 54** **(d) 81**

5.6	**(a)** $7^2 = 49$ $3 \times 49 = 147$ **(b)** $121 - 8 \times 10$ $= 121 - 80 = 41$ **(c)** $\sqrt{49} = 7$ $7 + 3 = 10$ $10 \times 10 = 100$ **(d)** $1 \times 4 = 4$ $5 + 4 = 9$ $9^2 = 81$	**(a) 147** **(b) 41** **(c) 100** **(d) 81**

Unit 2 · Estimation

Q	Working	Answer
estimating by rounding to 1 significant figure		
6.1	**(a)** $\frac{20 \times 30}{6} = \frac{600}{6} = 100$ **(b)** $\frac{50 \times 20}{2} = \frac{1000}{2} = 500$ **(c)** $\frac{60 \times 30}{6} = \frac{1800}{6} = 300$	**(a) 100** **(b) 500** **(c) 300**
6.2	**(a)** $\frac{40 \times 50}{2} = \frac{2000}{2} = 1000$ **(b)** $\frac{20 \times 30}{3} = \frac{600}{3} = 200$ **(c)** $\frac{80 \times 20}{8} = \frac{1600}{8} = 200$	**(a) 1000** **(b) 200** **(c) 200**
6.3	**(a)** $\frac{10 \times 60}{2} = \frac{600}{2} = 300$ **(b)** $\frac{30 \times 40}{6} = \frac{1200}{6} = 200$ **(c)** $\frac{50 \times 70}{7} = \frac{3500}{7} = 500$	**(a) 300** **(b) 200** **(c) 500**

6.4	**(a)** $\frac{30 \times 40}{4} = \frac{1200}{4} = 300$ **(b)** $\frac{70 \times 20}{2} = \frac{1400}{2} = 700$ **(c)** $\frac{40 \times 50}{4} = \frac{2000}{4} = 500$	**(a) 300** **(b) 700** **(c) 500**
6.5	**(a)** $\frac{90 \times 10}{2} = \frac{900}{2} = 450$ **(b)** $\frac{60 \times 40}{3} = \frac{2400}{3} = 800$ **(c)** $\frac{20 \times 50}{2} = \frac{1000}{2} = 500$	**(a) 450** **(b) 800** **(c) 500**
6.6	**(a)** $\frac{40 \times 20}{4} = \frac{800}{4} = 200$ **(b)** $\frac{70 \times 30}{7} = \frac{2100}{7} = 300$ **(c)** $\frac{50 \times 60}{5} = \frac{3000}{5} = 600$	**(a) 200** **(b) 300** **(c) 600**

Unit 2 · Factors and multiples problems

Q	Working	Answer
sum of factors and LCM problems		
7.1	**(a)** $1 + 2 + 4 + 8 + 16 + 32 = 63$ $4^3 - 1 = 63$ **(b)** $100 : 100, 200, 300, 400$ $80 : 80, 160, 240, 320, 400$	**(a) shown** **(b) 400 m**
7.2	**(a)** $1 + 2 + 5 + 10 = 18$ $3 \times 6 = 18$ **(b)** $40 : 40, 80, 120$ $30 : 30, 60, 90, 120$	**(a) shown** **(b) 120 m**
7.3	**(a)** $1 + 3 + 7 + 21 = 32$ $2^5 = 32$ **(b)** $50 : 50, 100, 150$ $75 : 75, 150$	**(a) shown** **(b) 150 m**

7.4	**(a)** $1+2+3+4+6+12=28$ $5^2+3=28$ **(b)** $120: 120, 240, 360$ $90: 90, 180, 270, 360$	**(a) shown** **(b) 360 m**
7.5	**(a)** $1+3+9=13$ $3^2+4=13$ **(b)** $24: 24, 48, 72$ $36: 36, 72$	**(a) shown** **(b) 72 cm**
7.6	**(a)** $1+2+4+8+16=31$ $2^5-1=31$ **(b)** $18: 18, 36, 54, 72, 90$ $30: 30, 60, 90$	**(a) shown** **(b) 90 cm**